Learning-based Control under Parametric Uncertainty

Abstract

1 Problem Formulation

Consider a control-affine system with parameteric uncertainty,

$$\dot{x}(t) = f(x(t)) + g(x(t))u(t) + Y(x(t))a \tag{1}$$

where $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the control input, $Y(x(t))$ is a matrix of known nonlinear features, and $a \in \mathbb{R}^p$ is a vector of unknown parameters. The goal is to design a neural network-based direct adaptive controller $u(t) = \pi(x(t), \hat{a}(t), r_a(t); \theta_\pi)$ such that the system is guaranteed to be stable under some Lyapunov function. The premise of this direct adaptive control design is that we are using a parameter estimation scheme to keep an estimation of the unknown parameter with a conservative error bound, i.e., find $\hat{a}(t), r_a(t)$ such that $a(t) \in B_{r_a(t)}(\hat{a}(t))$.

2 Proposed Approach

2.1 Parameter Estimation with Linear Observer Designs

Following the parameter estimator designs in [ADG09], we first build a state predictor $\hat{x}(t)$ and an associated filter $w(t)$:

$$\begin{aligned}
\text{predictor}: \quad & \dot{\hat{x}} = f(x) + g(x)u + Y(x)\hat{a} + ke + w\dot{\hat{a}} \quad , \quad e = x - \hat{x} \\
\text{filter}: \quad & \dot{w} = Y(x) - kw \quad , \quad w(t_0) = 0 \\
\text{prediction error dynamics}: \quad & \dot{e} = \dot{x} - \dot{\hat{x}} = \cancel{f(x) + g(x)u} + \underbrace{Y(x)a - (\cancel{f(x) + g(x)u}) - Y(x)\hat{a}}_{Y(x)\tilde{a}} - ke - w\dot{\hat{a}} \\
\text{Auxiliary signal}: \quad & \eta \triangleq e - w\tilde{a} \\
\text{above} \implies \quad & \dot{\eta} = \dot{e} - \dot{w}\tilde{a} + w\dot{\tilde{a}} \\
& = \cancel{Y(x)\tilde{a}} - ke - \cancel{w\dot{\hat{a}}} \quad \underbrace{-\dot{w}\tilde{a}}_{-\cancel{Y(x)\tilde{a}} + kw\tilde{a}} \quad + \cancel{w\dot{\hat{a}}} \\
& = -k(e - w\tilde{a}) = -k\eta \quad . \quad \eta(t_0) = e(t_0)
\end{aligned}$$

As a result, we obtain an auxiliary signal which is a linear observation of the unknown estimation error $\tilde{a}$: $e - y = w\tilde{a}$, where $y \to 0$ exponentially fast. Using this error term, we can use the same parameter update law in [ADG09] which guarantees estimation error is monotonically non-increasing:

$$\dot{\hat{a}} = \gamma w^T(e - \eta) \implies \frac{d}{dt}\tilde{a}^T\tilde{a} = -2\tilde{a}^T\gamma w^T(e - \eta) = -2\tilde{a}^T\gamma w^T w\tilde{a} \leq 0 \tag{2}$$

2.2 Direct Adaptive Control Design

Given a Lyapunov function $V(x)$, to ensure the system stay stable under the parametric uncertainty, one can describe the condition in the sense of "robust stability".

$$\max_{a(t) \in B_{r_a(t)}(\hat{a}(t))} \frac{\partial V}{\partial x}^T (f(x) + g(x)u + Y(x)a) = \max_{a(t) \in B_{r_a(t)}(\hat{a}(t))} \frac{d}{dt}V(x) \leq 0 \tag{3}$$

Note that the LHS in the above constraint is linear in a, meaning that there should always be a closed-form solution. Additionally, the constraint is also affine in u, which means we can write a closed-form projection that ensures u satisfies the stability constraint.

To deal with the scenarios where Lyapunov constraints might become restrictive, especially under the circumstances where there are control input saturation or underactutation in the system, we should also ultimately use a neural network to learn the Lyapunov function. But for experiments for now, we can start with simple prescribed ones like a quadratic function of x.

2.3 Learning Optimization Problem Setup

During training, we need to learn two neural networks, one for the control design, and one for the Lyapunov function. The learning process would be a self-supervised one, meaning that we optimize the weights of both networks to minimize a multi-trajectory loss. As an example, we can define the training loss as the trajectory tracking loss upon multiple different initial conditions and multiple trajectory snippets. We enforce the projection defined in the last subsection during training so the learned controller is certified to be stable based on the learned Lyapunov function.

3 Claimed Contributions

We would like to claim the following contributions:

- The proposed framework utilizes machine learning to circumvent the difficulties in designing adaptive controllers for underactuated systems where g might not be invertible.
- The proposed framework addresses the gap in the current literature where learning-based frameowrks are only certified with stability guarantees when the network approximation error is zero, ensuring learning-based controller is always safe.

References

[ADG09] Veronica Adetola, Darryl DeHaan, and Martin Guay. Adaptive model predictive control for constrained nonlinear systems. *Systems & Control Letters*, 58(5):320–326, 2009.